



Gaming motives, mental health, and belonging among Asian students in Canada and White Canadian university students: A cross-sectional study

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1. Introduction

Video gaming has become increasingly popular among young adults, with 53 % of Canadians gaming regularly, averaging almost 8 h per week [1]. Similarly, a large national cohort study reported that 70–80 % of Western young adults aged 18–24 played video games in the last year [2].

The effects of gaming on well-being appears to vary across studies and cultural contexts. Some research highlights higher rates of gaming disorder (GD) among Asian people [3,4], while other studies suggest that the risk may be more closely tied to cultural values, such as vertical individualism—characterized by a focus on independence and personal achievement [5]—in Western societies [6,7]. Furthermore, GD prevalence rates tend to be higher in Asian countries compared to Western nations [8], suggesting that Asian students may be at a higher risk for GD. Given these mixed findings on the cultural factors contributing to GD, the present study addresses disparities in gaming behaviour between Asian international/immigrant students and White Canadian students in Canada. For conciseness, we use the term “Asian students” to refer to Asian international or immigrant students residing in Canada throughout this manuscript. It particularly focuses on mental health outcomes (i.e., loneliness, anxiety, depression) and sociocultural factors (i.e., belongingness, social networks, and support systems). Accordingly, the present study explicitly aims to determine whether Asian international/immigrant and White Canadian university students differ in (a) gaming motives and problematic gaming, (b) the mental-health correlates of gaming, and (c) the sociocultural moderators (belonging, social support, and perceived gaming norms) of these links.

1.1. Problematic vs. non-problematic gaming

It is crucial to differentiate between non-problematic (normal) and problematic gaming, as they have markedly different impacts on mental health and psychosocial well-being [9,10]. Non-problematic gaming (i.e., playing video games without experiencing negative impacts such as

impaired psychological functioning, social difficulties, or academic problems) often provides benefits such as amusement, relaxation, and improved social interactions with family and friends [1,10,11]. On the other hand, problematic gaming—characterized by excessive gaming behaviour that includes elements such as cravings, loss of control, and negative consequences impacting psychological functioning, social relationships, or academic performance [10]—is linked to negative outcomes, like depression [12–14], anxiety [15], loneliness, and poor academic performance [15,16], among others.

Whereas some studies suggest that longer gaming duration is associated with more severe psychiatric symptoms, such as depression, anxiety, and ADHD [13,17–19], other research indicates that gaming duration alone does not reliably predict these symptoms [20,21]. Therefore, gaming motivation may be a more accurate differentiator between problematic and non-problematic gaming behaviour. Achievement-driven gaming, driven by the desire to master challenges and gain status, tends to correlate with problematic behaviour [22–24]. Furthermore, escapism, where individuals game to avoid real-life problems, introduces a more complicated dynamic. Though escapism can lead to positive outcomes such as stress relief and enjoyment [25–27], it is also linked to negative consequences like social isolation and reduced well-being [26,28], which can escalate to problematic gaming [22,23].

Excessive gaming escapism, especially when tied to neglecting responsibilities and social isolation, can exacerbate these mental health issues, leading to more severe outcomes like depression and loneliness [23,28]. The stress-coping model of problematic gaming supports this by showing how depression and social anxiety may motivate individuals to game as a means to escape negative states and fulfill their social needs online [29]. In this context, escapism not only provides temporary relief but may evolve into a maladaptive coping strategy leading to problematic gaming that mediates psychological factors like self-esteem, loneliness, and anxiety, ultimately leading to problematic gaming behaviour [20,22,30].

Gaming disorders have been officially recognized and classified

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within both international and national diagnostic frameworks, such as the *International Statistical Classification of Diseases* (11th ed.; ICD-11) [31] and *Diagnostic and Statistical Manual of Mental Disorders* (5th ed., text rev.; DSM-5-TR) [32]. These classifications underscore a growing acknowledgment of the significant impact that problematic gaming can have on mental health, personal relationships, and occupational functioning, reflecting a consensus within the medical and psychological communities about the potential harms associated with persistent gaming behaviour.

1.2. Social support and belongingness

Modern video games increasingly incorporate strong social elements, challenging the traditional view of gamers as isolated individuals [33]. In fact, many gamers actively engage in social interactions within these gaming environments, and online gaming communities have become important spaces for building both online and offline social connections [33,34]. Research shows that gamers who are socially engaged within these platforms tend to display fewer problematic gaming behaviour, whereas those who play in more solitary contexts often face reduced psychosocial well-being [34].

The social displacement hypothesis suggests that excessive online gaming can displace offline interactions, which are generally considered more beneficial for social skill development [35]. This displacement may lead to social deficits, as online interactions offer limited social support and closeness. In support of this view, social anxiety and greater reliance on online social support, along with fewer offline relationships, have been linked to problematic gaming [36]. Additionally, higher levels of online social support have been associated with increased depressive symptoms in gamers, whereas offline social support is typically linked to lower depressive symptoms [37]. This discrepancy indicates that online social interactions may not provide the same mental health benefits as offline ones, exacerbating depressive symptoms when they replace in-person relationships. Conversely, positive social connections within gaming communities can mitigate the negative effects of problematic gaming by fostering social support and appreciation, particularly for individuals with compromised social well-being who turn to intensive gaming as a coping mechanism [38]. Furthermore, playing online video games with friends can actually enhance offline social support and strengthen existing friendships [39], which is particularly valuable in post-secondary settings, students' sense of belonging—especially for international undergraduates—is closely tied to academic success and mental well-being [40]. A strong campus connection is associated with reduced suicidal ideation [41], and students with more social support are better protected against depression and excessive gaming [42].

1.3. Loneliness and social anxiety

Loneliness and social anxiety significantly impact mental and physical well-being within the gaming context. Online gaming offers socially anxious individuals a comfortable alternative to face-to-face interactions, potentially reducing social anxiety [43] but can also limit offline social experiences and lead to problematic gaming [12]. Studies reveal a positive correlation between weekly online gaming hours and social anxiety, with socially anxious individuals often turning to gaming as an escape, which may lead to negative outcomes [13,44].

Loneliness is also a strong predictor of pathological gaming [45] and GD [46–48], and when combined with social anxiety, it further exacerbates problematic gaming tendencies [19]. The negative consequences of gaming, such as sleep deprivation and relationship difficulties, are intensified by loneliness [49]. During the COVID-19 lockdowns, loneliness and the prevalence of mental health problems rose drastically compared to pre-pandemic levels [50,51]. These heightened levels of loneliness, depression, and anxiety led many young adults to intensify their gaming habits, amplifying the effects of loneliness and social

anxiety on gaming behaviour [52].

1.4. Cultural differences and international students

Gaming addiction is a growing concern in Asia, with countries like China and South Korea recognizing it as a public health issue and implementing prevention measures [32,53]. Studies show a higher prevalence of GD in Asia (5.08 %) compared to Europe (2.72 %) and North America (2.74 %) [8]. Factors such as the popularity of esports, gaming cafes, and permissive cultural attitudes contribute to this trend [8,21,53]. However, cross-cultural research shows mixed results and while some suggest Asians have higher rates of disordered gaming than Whites [3,4], other research points to higher susceptibility in Western cultures characterized by individualism, where gaming escapism is linked to negative outcomes, including GD [6,7,23,54].

Cultural norms significantly shape how individuals cope with stress. East Asian cultures, which prioritize harmony and indirect problem-solving, tend to favor coping strategies like avoidance and patience [55–57]. In contrast, Western cultures emphasize personal agency and autonomous problem-solving, aligning with a focus of individualism [58]. This contrast reflects broader cultural paradigms of collectivism in Asia versus individualism in the West. Empirical evidence shows that Asian Americans are more inclined to use avoidance strategies [59], while immigrant college students in the US demonstrate shifts based on acculturation levels, with first-generation immigrants preferring collective coping, and later generations adopting more active strategies common in Western cultures [58,60,61]. These findings underscore the intricate interplay between cultural heritage, acculturation, and coping behaviour, offering insight into how diverse populations manage stress and adversity.

Canadian universities have experienced a significant rise in international student enrollments, rising by 286 % over the past decade [62], with 21.2 % of post-secondary students in Ontario being international [63]. This growth underlines the need for research on international students, particularly in areas like mental health, sociocultural factors, and gaming. International students face unique challenges, including acculturation stress, language barriers, and social isolation, which contribute to issues like anxiety, depression, and social anxiety [64,65]. These issues are especially pronounced among Asian international students due to the cultural clash between interdependent, harmony-focused norms of their home countries and the more individualistic values of Western societies [58,66,67]. The resulting cultural dissonance intensifies feelings of stress, depression, social anxiety, and loneliness [56,68–70]. Social support is critical in mitigating these effects, with higher levels of belonging and community leading to better mental health outcomes and student retention [65,71]. However, international students often report lower levels of social support [72,73], compounding their mental health challenges [64]. The COVID-19 pandemic has further exacerbated these issues, leading to increased loneliness and gaming behaviour [74]. Understanding the cultural dynamics of acculturative stress is vital to effectively supporting the mental health of international students.

Gaming culture has also been profoundly shaped by the rise of competitive gaming (i.e., esports), which carries mental health implications and cultural nuances. In recent years, esports has grown into a global phenomenon with professional leagues and collegiate teams, making competitive gaming a mainstream activity for youth and university students [75]. This trend is especially pronounced in parts of Asia, where professional gaming is highly popular and culturally valued. While participation in esports can offer benefits—such as teamwork, social interaction, and a sense of achievement—studies warn that the intensive training and prolonged play associated with esports may exacerbate stress and mental health risks [75]. A systematic review reported that excessive competitive gaming is linked to issues like depression, anxiety, sleep disturbances, and social withdrawal among players [75]. Nonetheless, emerging evidence suggests a more nuanced

perspective: moderate involvement in esports or online gaming communities can enhance social connections and even help reduce stress [76,77]. Cultural attitudes toward esports differ, with East Asian countries normalizing professional gaming to a greater extent than Western contexts [8,21,53]. These cultural variations in esports engagement highlight the importance of considering competitive gaming environments when comparing gaming habits and mental health across student populations.

Beyond individual pathology, video gaming plays a significant social role in student life that can influence belonging, identity, and integration. Many university students, including international students, turn to gaming communities as a way to find peer support and a sense of belonging on campus. Online multiplayer games and campus gaming clubs often serve as social hubs where students form friendships and express shared identities [35]. Indeed, research indicates that online games can be socially advantageous for shy or socially anxious individuals, enabling them to overcome face-to-face communication barriers and build new friendships in a comfortable virtual space [35,43]. Gaming can thus become part of a student's identity, providing membership in a global "gamer" culture that transcends language and geographical boundaries. Recent qualitative work among Chinese students, for example, found that a popular multiplayer game had become "an indispensable tool for socializing," helping players maintain friendships and feel more integrated in their new environment [78]. In this way, digital gaming is not merely an isolating pastime as it is intertwined with students' social networks and cultural adjustment. These social and cultural dimensions of gaming are especially relevant for international students, who may leverage online games to cope with loneliness or to connect with co-nationals and local peers. Recognizing the positive facets of gaming, such as community-building and identity exploration, alongside its risks is crucial for a balanced understanding of how gaming affects student well-being.

1.5. Theoretical framework and implications of cross-cultural comparison

Cross-cultural frameworks predict that the meaning of gaming—and its links to mental health—will vary with culturally patterned selves, norms, and migration experiences. First, gaming studies indicate that cultural orientations such as individualism–collectivism and vertical individualism shape why people play and how those motives relate to disordered gaming. In a cross-national comparison, collectivism was linked to stronger social motives that, in turn, were associated with GD among Singaporean young adult gamers. In contrast, vertical individualism was linked to competitive motives and GD among Australian young adult gamers [79]. Relatedly, vertical individualism has been shown to strengthen risk pathways in gaming by amplifying the association between inattention and GD in young adult gamers [7]. Second, acculturation perspectives emphasise that international and immigrant students navigate acculturative stressors (e.g., language, norms, social networks) that affect both psychological and sociocultural adaptation [80,81]. These processes can amplify reliance on coping or recreational motives, and may alter the strength/direction of motive–symptom associations when belonging and support are low. This emphasis aligns with gaming research that treats motives as proximal mechanisms linking context to adjustment, including work showing that coping-oriented motives are often associated with higher GD symptoms [22,23].

Third, cross-national differences in cultural tightness–looseness (strength of norms and tolerance for deviance) imply that perceived gaming norms (approval/acceptance) will condition how motives link to outcomes. In tighter normative climates, deviations (e.g., intensive play) may carry stronger social meaning, potentially intensifying associations with symptoms—or, conversely, normatively approved play may be less distressing [82]. Finally, compensatory/stress–coping models of online behaviour propose that some individuals use games to regulate negative affect or unmet social needs [22,28,83]. When gaming is used as avoidance coping, this can be associated with poorer well-being.

Motives are central in these accounts, and meta-analytic evidence shows robust links between motives, especially escape/coping, and symptoms of gaming disorder. Although our moderators are perceived norms rather than value orientations, prior gaming work showing that vertical individualism can intensify risk pathways supports the broader expectation that cultural context moderates links between play-related factors and symptoms [7,79].

Comparing Asian students in Canada (international/immigrant) and White Canadian students tests whether cultural orientations evidenced in gaming research, acculturation contexts, and perceived norms shift (a) group averages in specific motives, and (b) the strength of motive–symptom links as a function of belonging, support, and norms. This moves beyond prevalence differences to a theory-driven analysis of when and for whom gaming motives relate to mental health. Our approach is consistent with the gaming-specific evidence above while acknowledging that we did not measure cultural value orientations directly.

1.6. The present study

Given the widespread engagement in gaming across Western countries and growing concerns about its mental health implications, we sought to investigate gaming disparities between Asian international/immigrant students and White Canadian students. We specifically analyzed differences in mental health outcomes, such as loneliness, anxiety, and depression. In addition, we examined the relationship between gaming behaviour and these mental health outcomes to understand whether gaming predicts or exacerbates these issues. In a separate analysis, we examined the influence of sociocultural factors on well-being, including students' sense of belongingness and the level of social support they receive from their environment. The study was motivated by findings that gaming-related disorders are more prevalent in Asian countries than in the West [8].

To account for potential confounding factors, we controlled for sociodemographic factors such as gender, age, years lived in Canada, and international status. Research has shown that males report higher rates of both non-problematic and problematic gaming compared to females [8,86]. The length of residence in Canada was also controlled for to account for its potential influence on gaming behaviour and related mental health and sociocultural outcomes. By controlling for these variables, the study aimed to reduce the potential confounding effects of sociodemographic differences on the observed gaming disparities between Asian international/immigrant and White Canadian students.

The objectives of this study were:

To examine the extent to which differences in gaming behaviour between Asian international/immigrant and White Canadian students exist. We hypothesized that Asian students would exhibit significantly different gaming motives and problematic gaming behaviour compared to White Canadian students (H1) [6,8].

To analyze the extent to which mental health outcomes (i.e., loneliness, anxiety, depression, and social anxiety) differ between these two groups in relation to their gaming behaviour. We hypothesized that mental health outcomes would significantly differ between Asian and White students in relation to their gaming behaviour (H2) [30,38].

To explore disparities in sociocultural factors (i.e., belongingness, online social support, offline social support, and gaming-related norms) between Asian international/immigrant and White Canadian students and investigate how these factors influence their gaming behaviour. We hypothesized that Asian international/immigrant and White Canadian students would exhibit significant differences in sociocultural factors, such as belongingness, online social support, offline social support, and gaming-related norms, which would influence their gaming behaviour (H3) [40,41].

2. Material and methods

2.1. Participants

We recruited university students aged 17–24, including Asian international/immigrant and White Canadian students at various academic levels (Undergraduate, Master's, Doctorate). Eligibility criteria included proficiency in English reading and engagement in video gaming. The target sample size was 216 participants, with an even distribution between the two groups (108 Asian and 108 White students). This number was derived from a power analysis using G*Power (version 3.1.9.7) for linear multiple regression [87,88], with an effect size of 0.15, an α of 0.05, power set at 0.95, and 19 predictors. The sample consisted of 247 university students ($M = 19.7$, $SD = 1.70$). Of these participants, 42.5 % ($n = 105$) self-identified as female. The racial/ethnic composition of the recruited sample is divided between 137 White (55.5 %) and 110 Asian (44.5 %) students. Among Asian participants ($n = 110$), 53 (48.2 %) were international students, 54 (49.1 %) were first- or second-generation immigrants, and 3 (2.7 %) were third-generation (or later) immigrants. Additional demographic information is presented in Table 1 in the Appendix.

We conducted this study from January to July 2024, recruited participants from a university in Ontario, Canada and social media channels. Additional recruitment targeted specific groups, such as Asian student associations and university courses (with instructors' consents).

2.2. Procedure

The online Qualtrics survey, estimated to take 45 min, explored gaming behaviour and mental health outcomes, including loneliness, anxiety, social anxiety, and depression, as well as sociocultural factors, such as belongingness and social support. The survey was conducted in English, to maintain timeline constraints and data consistency.

The flexibility of the online survey format via Qualtrics allowed participants to complete it at their convenience. Prior to the start of the survey, participants were presented with an electronic informed consent form detailing the study's objectives, procedures, potential risks, benefits, and confidentiality measures. Consent was documented through checkboxes at the end of the forms, with disagreement redirecting participants to a thank-you webpage and exiting the survey. Upon providing consent, participants first encountered a series of socio-demographic questions, followed by items related to their gaming behaviour. Following this, participants answered a series of questions regarding their mental health and sense of belongingness. Participants then responded to questions regarding social networks, both online and offline, and gaming norms. Finally, they reached a thank-you page, concluding the survey.

Compensation for participation included academic credit who registered through the university portal as part of their coursework, and a chance to win one of several Amazon gift cards. The survey incorporated an attention-check question to ensure the validity of responses. The question was "To show that you are paying attention, please select 'Several days.'" Participants retained eligibility for the draw even if they withdrew from the study. Email addresses for the draw were collected at the survey's end.

Among 153 White respondents, 152 completed the survey (99.35 % completion; 0.65 % break-off), and among 139 Asian respondents, 137 completed (98.56 % completion; 1.44 % break-off). Overall completion exceeded 98 %, minimizing concerns about non-response bias.

2.3. Measures

2.3.1. Sociodemographic characteristics

The sociodemographic section collected key information, including age, academic level, faculty, program year, enrollment status, international student status, gender identity, sexual orientation, Indigenous

identity, place of birth, time in Canada, primary home language, ethnic/racial background, refugee status, family composition, parents' birthplace and education, household income, and living arrangements during studies.

2.3.2. Social anxiety symptoms

Social phobia symptoms were measured using the Short Form Social Interaction Anxiety (SIAS-6) [89,91], a six-item self-report scale developed to assess social anxiety. Participants rated each item on a 5-point Likert-type scale from 0 (*Not at all*) to 4 (*Extremely*), with higher scores indicating greater social phobia symptoms. The SIAS-6 has shown high internal reliability with a McDonald's ω of 0.82 [90] and Cronbach $\alpha = 0.84$ [92], demonstrating strong convergent validity across Western and Asian samples [90,92]. It is useful for screening, clinical treatment outcomes, and assessing social phobia severity in various studies, demonstrating good internal consistency in our sample as well ($\alpha = 0.84$).

2.3.3. Loneliness symptoms

The revised UCLA Loneliness Scale is a 20-item self-report measure developed to assess feelings of loneliness and satisfaction with social relationships [93]. Participants rated each item on a 4-point Likert-type scale from 1 (*Never*) to 4 (*Often*). Higher scores on the scale indicate a greater frequency of experiencing loneliness and dissatisfaction with social relationships. The scale has demonstrated high internal consistency with Cronbach α of 0.94, indicating its reliability. The revised UCLA Loneliness Scale is useful for assessing loneliness in research and clinical settings and showed good internal consistency in our sample ($\alpha = 0.79$).

2.3.4. Generalized anxiety symptoms

The Generalized Anxiety Disorder 7-item scale (GAD-7) is a brief self-report measure designed to assess the severity of generalized anxiety disorder symptoms in clinical practice and research [94]. It consists of 7 items, each scored from 0 (*Not at all sure*) to 3 (*Nearly every day*), with higher scores indicating more severe anxiety symptoms. The GAD-7 demonstrated excellent internal consistency with Cronbach's α of 0.89, 0.91, and 0.92 [94–96], indicating high reliability. This scale is a valuable tool for screening and assessing the severity of generalized anxiety disorder due to its strong psychometric properties and clinical utility. It demonstrated good internal consistency in our sample ($\alpha = 0.87$).

2.3.5. Depression symptoms

The shortened depression scale used in this study was the 10-item Center for Epidemiologic Studies Depression Scale (CES-D 10) [97]. The CES-D 10 is a self-report scale developed to assess depression symptoms. Participants were asked to rate the frequency of experiencing each item over the past week on a 4-point Likert-type scale ranging from 0 (*Rarely or none of the time*) to 3 (*Most or all of the time*). Higher scores on the items represented a higher frequency of the occurrence of depression symptoms. The CES-D 10 demonstrated good internal consistency with a Cronbach's alpha of 0.88. The scale also showed good agreement with the 20-item CES-D scale, with a κ of 0.82. The CES-D 10 has been found to be a reliable and valid measure of significant depression symptoms among HIV-positive participants and showed good internal consistency in our sample ($\alpha = 0.80$).

2.3.6. Substance use

The Ontario Child Health Study (OCHS) [98] substance use measure assesses cigarette, cannabis, and alcohol consumption. For cigarettes, the measure determines past use, frequency of use in the last six months, and average daily consumption. Cannabis use is gauged by recent use, past use, and frequency of consumption. Alcohol use is measured by frequency and regularity of consumption within the past six months.

2.3.7. University belongingness

University belongingness was assessed using The College Belongingness Questionnaire (CBQ) [99] and the staff relations subscale of the University Belongingness Questionnaire (UBQ) [100]. The CBQ is a 10-item self-report scale developed to assess college students' sense of belonging [99]. Participants rate their agreement with each item on a 7-point Likert-type scale, ranging from 1 (*Strongly disagree*) to 7 (*Strongly agree*). Higher scores indicate a greater level of college belongingness. The CBQ demonstrates strong internal reliability estimates with Cronbach's $\alpha = 0.79$ – 0.93 and has been found to be psychometrically reliable and valid for assessing undergraduate students' sense of belonging, as evidenced by good internal consistency in our sample ($\alpha = 0.90$). The UBQ is a 24-item self-report scale developed to assess university belonging [100]. The scale has three subscales: university affiliation, university support and acceptance, and faculty and staff relations. Only the staff relations subscale is used in our study. Participants were asked to rate the extent to which each item applied to them on a 4-point Likert-type scale ranging from 1 (*Strongly disagree*) to 4 (*Strongly agree*). Higher scores on the items represented a higher level of university belonging. The UBQ has shown good internal consistency, with Cronbach's α of 0.93 for the total score and 0.88 for faculty and staff relations factors. The scale has also demonstrated good convergent and incremental validity, positively correlating with measures of perceived social support, social connectedness, and general belonging. It demonstrated good internal consistency in our sample ($\alpha = 0.86$).

2.3.8. Problematic gaming

Problematic online gaming was assessed using the 12-item Problematic Online Gaming Questionnaire Short-Form (POGQ-SF) [101]. The POGQ-SF is a 12-item scale measuring six dimensions of problematic online gaming: preoccupation, overuse, immersion, social isolation, interpersonal conflicts, and withdrawal. Participants rate their agreement with each item on a 5-point Likert-type scale, ranging from 1 (*Never*) to 5 (*Always*). Higher scores indicated increased problematic gaming behaviour. The POGQ-SF has shown high internal reliability with a Cronbach's α of 0.91, indicating its consistency. The scale has been found to be a valuable tool for assessing problematic online gaming in both survey research and clinical settings. It showed good internal consistency in our sample ($\alpha = 0.88$).

2.3.9. Gaming motives

The Videogaming Motives Questionnaire (VMQ) [102] was employed to assess gaming motives. It is a 24-item self-report scale developed to assess eight gaming motivations: recreation, social interaction, coping, violent reward, fantasy, cognitive challenge, customization, and competition. Participants are asked to rate the extent to which they agree with each item on a 5-point Likert-type scale ranging from 0 (*Strongly disagree*) to 4 (*Strongly agree*). Higher scores on the items represent a higher frequency of the occurrence of the corresponding gaming motive. The VMQ has demonstrated strong psychometric properties, including good to excellent internal consistency with Cronbach α of 0.91 and acceptable test–retest scores with all values higher than 0.60. The VMQ has been found to be useful in identifying individual differences in gaming behaviour. This scale showed good internal consistency across its factors in our sample (mean $\alpha = 0.84$).

2.3.10. Online and offline social support and gaming-related norms

Two subscales of the Online Gaming Social Networks and Distinctive Experiences Survey [103] were used to assess social support (online and offline) and gaming-related norms. The Offline/Online Social Support subscale consists of two 7-item scales to evaluate offline and online social support, which were adapted from the Interpersonal Support Evaluation List. The offline social support scale aimed to gauge the degree to which participants experienced offline companionship, trusted advice, and social interaction with family and friends. Similarly, the online social support scale sought to assess the extent to which participants

experienced online companionship, trusted advice, and social interaction with family and friends. In both scales, participants were required to indicate their level of agreement with each item based on their experiences over the past year, using a 5-point Likert scale ranging from 1 (*Strongly disagree*) to 5 (*Strongly agree*). Higher scores on the online/offline social support scale indicate a greater level of perceived social support in the respective domain. Both the online ($\alpha = 0.73$) and offline ($\alpha = 0.79$) social support subscales showed good internal consistency in our sample. Additionally, the Gaming-Related Norms subscale is a 10-item measure to examine participants' perceptions of societal attitudes toward gaming. This measure aimed to capture the extent to which participants believed that important social and cultural institutions, as well as other individuals such as family members and friends, held positive judgments about gaming. Participants rated these perceptions on a 5-point Likert scale ranging from 1 (*Strongly disagree*) to 5 (*Strongly agree*), with higher scores indicating a greater perception of positive societal attitudes toward gaming. The measure showed good internal consistency in our sample ($\alpha = 0.81$).

2.4. Data analysis

The data were analyzed using R (Version 4.4.1) [104]. Preparatory steps included data cleaning to remove incomplete responses, attention check failures, and participants outside the 17–24 age range or who did not identify as Asian (Southeast/East Asian) or White, including mixed Asian-White identities. We also screened the data for normality, checking skewness and kurtosis, to ensure the appropriateness of subsequent analyses.

For H1, a series of linear regressions were conducted to assess the effect of ethnicity on gaming behaviour, including problematic, recreational, competitive, cognitive, coping, social, violence, customization, fantasy motives, and gaming norms. Each model controlled for age, gender (dummy-coded), years in Canada (categorized and dummy-coded), and international student status to isolate ethnicity's unique effect.

To test H2, moderation analyses were performed to evaluate the interaction effects between ethnicity and gaming behaviour on mental health outcomes (loneliness, anxiety, depression, and social anxiety), controlling for the same sociodemographic variables.

For H3, complex moderation analyses were conducted, regressing mental health outcomes on gaming behaviour, sociocultural factors (belongingness, online and offline social support, and gaming norms), and their interaction with ethnicity. Control variables included age, gender, years in Canada, and international student status. These analyses aimed to reveal how sociocultural factors moderate the relationship between gaming behaviour and mental health across the two ethnic groups.

3. Results

Descriptive statistics for all study variables by ethnicity are presented in Table 1 in the Appendix. Asian students reported higher problematic gaming and higher social/violent motives. In adjusted regressions, only violent and social motives remained significantly higher among Asian students; problematic gaming did not differ by group after covariates. Subsequent models examined whether links between motives and depression/loneliness/anxiety varied by ethnicity and by belonging, social support, and gaming norms (two- and three-way interactions).

3.1. Objective 1: differences in gaming behaviour between Asian and White students

Linear regression analyses were conducted to examine differences in gaming motives between Asian and White students and accounted for a statistically significant and moderate proportion of variance ($R^2 = 0.17$, $F(9, 233) = 5.43$, $p < 0.001$, $R_{adj}^2 = 0.14$). Ethnicity was a significant

predictor, with Asian students reporting higher violent gaming motives than White students ($B = 1.32$, 95 % CI [0.26, 2.38], $t(233) = 2.45$, $p = 0.015$). Similarly, a second model predicted social gaming motives and explained a significant, though smaller, proportion of variance ($R^2 = 0.10$, $F(9, 233) = 2.75$, $p = 0.005$, $R^2_{adj} = 0.06$). Again, ethnicity was a significant predictor, with Asian students reporting higher social gaming motives than White students ($B = 1.32$, 95 % CI [0.19, 2.46], $t(233) = 2.29$, $p = 0.023$). These findings suggest that Asian students are more motivated by violent and social aspects of gaming compared to their White counterparts. Nonsignificant results are excluded for brevity.

3.2. Objective 2: extent to which mental health outcomes differ in relation to gaming behaviour

Regression analyses were performed to assess whether the relationship between gaming motives and mental health outcomes differed by ethnicity. In the first model predicting depressive symptoms, the interactions between ethnicity and recreational gaming motives were significant ($B = 0.06$, 95 % CI [0.02, 0.10], $t(231) = 2.76$, $p = 0.006$). The model explained a significant proportion of variance ($R^2 = 0.10$, $F(11, 231) = 2.33$, $p = 0.010$, $R^2_{adj} = 0.06$), indicating that recreational gaming motives were more strongly associated with depressive symptoms among Asian students than among White students. Similarly, the interaction between ethnicity and coping gaming motives on depressive symptoms was significant ($B = 0.58$, 95 % CI [0.02, 1.13], $t(231) = 2.05$, $p = 0.041$). This model accounted for a significant proportion of variance ($R^2 = 0.09$, $F(11, 231) = 1.96$, $p = 0.033$, $R^2_{adj} = 0.04$), suggesting that coping motives were more strongly related to depressive symptoms in Asian students. For loneliness, significant interactions were found between ethnicity and recreational gaming motives ($B = 0.04$, 95 % CI [0.007, 0.06], $t(231) = 2.42$, $p = 0.016$). Although the model explained a smaller proportion of variance ($R^2 = 0.06$, $F(11, 231) = 1.26$, $p = 0.248$, $R^2_{adj} = 0.001$), the results indicated that recreational gaming motives are more strongly associated with loneliness among Asian students. Similarly, the interaction between ethnicity and coping gaming motives on loneliness was significant ($B = 0.51$, 95 % CI [0.11, 0.91], $t(231) = 2.52$, $p = 0.012$). This suggests that coping motives are more strongly related to feelings of loneliness among Asian students.

3.3. Objective 3: sociocultural factors influencing gaming behaviour and mental health

Three-way interactions were examined to explore how sociocultural factors influence the relationship between gaming motives and mental health outcomes across ethnic groups.

The interaction among ethnicity, recreational gaming motives, and gaming-related norms on depressive symptoms was significant ($B = 0.007$, 95 % CI [0.001, 0.013], $t(227) = 2.29$, $p = 0.023$). The model accounted for a significant proportion of variance ($R^2 = 0.12$, $F(15, 227) = 2.12$, $p = 0.010$, $R^2_{adj} = 0.07$). This indicates that gaming norms moderate the relationship between recreational gaming motives and depression differently for Asian and White students, with a stronger effect observed in Asian students. A significant three-way interaction was also found among ethnicity, competition gaming motives, and sense of belonging on depressive symptoms ($B = 0.04$, 95 % CI [0.006, 0.08], $t(227) = 2.31$, $p = 0.022$). This model explained a moderate proportion of variance ($R^2 = 0.23$, $F(15, 227) = 4.61$, $p < 0.001$, $R^2_{adj} = 0.18$), suggesting that the sense of belonging strengthens the association between competition motives and depression more in Asian students than in White students. Additional analyses revealed significant interactions for cognitive gaming motives and sense of belonging on depressive symptoms ($B = 0.05$, 95 % CI [0.005, 0.09], $t(227) = 2.21$, $p = 0.028$). The model accounted for a significant proportion of variance ($R^2 = 0.25$, $F(15, 227) = 4.96$, $p < 0.001$, $R^2_{adj} = 0.20$), indicating a stronger moderating effect of belonging for Asian students. Furthermore, a significant three-way interaction was found between ethnicity, coping

gaming motives, and sense of belonging on depressive symptoms ($B = -0.04$, 95 % CI [-0.08, -0.007], $t(227) = -2.36$, $p = 0.019$). The model explained a moderate proportion of variance ($R^2 = 0.25$, $F(15, 227) = 4.98$, $p < 0.001$, $R^2_{adj} = 0.20$) suggesting that, for Asian students, a higher sense of belonging mitigates the relationship between coping motives and depression. In terms of loneliness, significant interactions were observed between ethnicity, violent gaming motives, and sense of belonging ($B = 0.03$, 95 % CI [0.010, 0.05], $t(227) = 3.00$, $p = 0.003$). The model explained a moderate proportion of variance ($R^2 = 0.36$, $F(15, 227) = 8.42$, $p < 0.001$, $R^2_{adj} = 0.32$), suggesting that, for Asian students, at high levels of violent gaming motives, sense of belonging is less effective at reducing loneliness. Significant interactions were also found between ethnicity, recreational gaming motives, and gaming norms on social anxiety ($B = 0.006$, 95 % CI [0.001, 0.012], $t(227) = 2.19$, $p = 0.029$). The model accounted for a weak but significant proportion of variance ($R^2 = 0.10$, $F(15, 227) = 1.73$, $p = 0.046$, $R^2_{adj} = 0.04$). This suggests that gaming norms influence the relationship between recreational motives and social anxiety differently across ethnic groups, with a stronger association for Asian students. Finally, significant interactions were found between ethnicity, cognitive gaming motives, and online social support on social anxiety ($B = -0.008$, 95 % CI [-0.015, -0.001], $t(227) = -2.17$, $p = 0.031$). This indicates that online social support moderates the relationship between cognitive gaming motives and social anxiety differently for Asian students, with a weaker association observed.

4. Discussion

The present study investigated differences in gaming behaviour, mental health outcomes, and the moderating role of sociocultural factors among Asian international/immigrant students and White Canadian students. Our findings revealed that Asian students scored significantly higher on violent and social gaming motives compared to their White counterparts. This suggests that Asian students are more motivated by the violent and social aspects of gaming, which aligns with some prior research indicating cultural differences in gaming preferences [21]. Collectivist cultures, such as those in many Asian societies, generally report lower overall aggression rates compared to individualistic cultures [105]. However, the impact of violent media is moderated by cultural attitudes and norms surrounding aggression. In collectivist societies, overt aggression is often culturally suppressed, leading individuals to seek indirect means of expressing or managing aggressive impulses. Violent video games may serve as a socially acceptable outlet for channelling aggression [106], aligning with cultural norms that discourage explicit expressions of aggression. Additionally, repeated exposure to violent gaming can reinforce aggressive cognitive frameworks even within collectivist cultures, suggesting that higher violent gaming motives among Asian students may help them navigate and internalize aggressive thoughts within a culturally constrained environment. The emphasis on social aspects of gaming among Asian students can also be linked to the collectivist value of social harmony and group cohesion, reflecting a desire to engage in cooperative or competitive interactions that are culturally reinforced [107,108]. Consistent with this, cross-national work in young adult gamers shows that collectivism predicts stronger social motives, which in turn relate to GD among Singaporean gamers, whereas vertical individualism predicts competitive motives and GD among Australian gamers [79]. Our higher social motives for Asian students, therefore, converge with the idea that culture shapes motives.

The observed differences in violent and social gaming motives are noteworthy, as few studies (to our knowledge) have explored specific gaming motives across cultural groups. This contrasts with previous research that often focused on overall gaming behaviour or problematic gaming without investigating the specific motivations behind gaming activities. Our findings contribute new insights into how cultural factors may influence gaming preferences, highlighting the importance of

considering specific motives rather than general gaming behaviour.

Contrary to some studies that reported higher problematic gaming among Asian populations in Western contexts [3,4], we found no significant differences in problematic gaming behaviour between the two groups. This also contrasts with research suggesting that Western individuals are more prone to problematic gaming [6,7]. The discrepancy may stem from factors such as cultural assimilation, differences in gaming environments, or varying definitions and measurements of problematic gaming across studies [25]. These factors can influence how problematic gaming is experienced and reported across different cultural contexts.

Our results indicated that the association between gaming behaviour and mental health outcomes was more pronounced for Asian students, especially regarding coping and recreational gaming motives, which were more strongly linked to depressive symptoms and loneliness. Specifically, Asian students who engaged in gaming for recreational purposes or as a coping mechanism experienced higher levels of depression and loneliness compared to their White counterparts. The stronger associations observed among Asian students may stem from cultural differences in coping strategies. Asian cultures often favour avoidance and internally focused coping mechanisms in response to stress [55–57,59]. Consequently, Asian students may be more likely to use gaming as a means of escape or coping, which can exacerbate mental health issues like depression and loneliness [23,28]. This contrasts with some earlier research suggesting that escapism through gaming was more harmful for Western populations [54], highlighting the need to consider cultural context when examining gaming behaviour. These patterns are consistent with motives functioning as proximal mechanisms through which cultural context influences maladaptive outcomes [79].

Sociocultural factors played a significant moderating role in these relationships. The sense of belonging and gaming-related norms were found to moderate the relationship between gaming motives and mental health outcomes more strongly for Asian students. For instance, gaming norms strengthened the association between recreational gaming and depression more in Asian students, indicating that cultural expectations around gaming may intensify its impact on mental health. This moderation pattern conceptually echoes findings that vertical individualism can amplify risk pathways in gaming: vertical individualism strengthened the link from inattention to GD in young adult gamers [7]. Although our moderators were belonging and perceived norms rather than value orientations or inattention, both studies indicate that culturally relevant context can intensify links between proximal risk factors (e.g., motives or symptoms) and adverse outcomes.

Conversely, a higher sense of belonging mitigated the negative impact of using gaming as a coping mechanism on depression for Asian students. This underscores the protective role of belongingness in buffering against mental health issues among Asian international/immigrant students [40,65,71]. In terms of loneliness, sense of belonging moderated the relationship between violent gaming motives and loneliness more strongly for Asian students. Fostering a strong sense of community and belonging may help alleviate feelings of loneliness associated with certain gaming behaviour among Asian students. These findings collectively emphasize the crucial role of social support in collectivistic cultures, where interpersonal relationships and community connections are central to individual well-being [5,67]. While online gaming can provide an escape from distress, excessive use can displace offline face-to-face experiences [12,35]. Our results suggest that offline social support networks are more effective in mitigating negative mental health outcomes for Asian students. The weaker effect of online social support indicates that digital interactions may not fully substitute for the benefits derived from in-person social connections in collectivist cultures.

Our findings highlight the complex interplay between cultural factors, gaming behaviour, and mental health outcomes. The stronger associations observed among Asian students may be due to the unique

challenges they face, such as acculturative stress, language barriers, and social isolation [64,70]. Additionally, cultural differences in coping strategies may influence how gaming affects mental health. Asian cultures often favour avoidance and internally targeted control strategies in response to stress [55,59], which may lead individuals to use gaming as a means of coping, potentially exacerbating negative mental health outcomes.

Asian students in Canada reported higher social and violent gaming motives but did not differ from White Canadian peers in adjusted levels of problematic gaming. Thus, gaming behaviour was similar across groups, while psychological cost per motive was higher for Asian students. However, recreational and coping motives were more strongly associated with depressive symptoms and loneliness among Asian students. Sociocultural context further shaped these relationships. In more permissive gaming climates, recreational motives predicted sharper increases in depressive symptoms and social anxiety. A strong sense of belonging buffered the coping–depression link yet amplified the competition–depression and cognitive–depression associations, and it became less protective against loneliness when violent-play motives were high. Finally, greater online social support weakened the cognitive–motive association with social anxiety for Asian students. This pattern aligns with self-construal accounts in which motivation and emotion regulation are socially embedded, acculturation perspectives highlighting stressors for international and immigrant students, and tightness–looseness differences in normative climates. It also fits compensatory and stress-coping models that locate risk in why people play rather than how long, consistent with meta-analytic links between escape/coping motives and disordered play.

Mitigating the mental health impacts of problematic gaming among students will require evidence-based, culturally informed strategies. Fortunately, a growing literature points to effective interventions for gaming-related problems. Recent reviews of clinical trials indicate that cognitive-behavioural approaches (both individual and group-based) can significantly reduce gaming disorder symptoms [84]. For instance, structured group counselling and short-term CBT have been shown to help student gamers regain control over gaming and alleviate associated depression or anxiety. Beyond formal therapy, prevention and harm-reduction initiatives are increasingly emphasizing education and self-regulation [8]. Gamers tend to respond favorably to strategies that support informed decision-making and autonomy (e.g., psychoeducational workshops, self-monitoring apps, or healthy gaming guidelines), rather than punitive measures. A key opportunity lies in leveraging the protective factors identified in our study. Facilitating strong social support and campus belonging can directly buffer students against the loneliness and stress that often drive escapist gaming. Universities might consider peer support groups or student-led gaming clubs as outlets for at-risk gamers to form positive connections. Such programs align with evidence that supportive gaming communities can foster well-being. For example, casual esports participation has been linked to stress relief and enhanced social connectedness [76,77]. In sum, a multifaceted approach that combines therapeutic intervention (targeting maladaptive gaming motives) with campus-based prevention (promoting healthy gaming habits and social inclusion) may best mitigate the mental health risks of problematic gaming in international and domestic students alike.

Future research should build on motive-based and cultural frameworks to deepen our understanding of gaming and mental health. Longitudinal studies are needed to clarify the directionality between gaming motives and psychological outcomes, as our cross-sectional design cannot determine causality. For example, tracking students over time could reveal whether certain motives (like gaming to escape stress) predict later increases in depression, or if worsening mental health in turn intensifies escapist gaming, or both. It would also be valuable to extend this work to diverse cultural groups. The current study focused on Asian international/immigrant and White Canadian students, but “Asian” students are themselves a heterogeneous category. Future

studies could compare subgroups (e.g. East Asian vs. Southeast Asian backgrounds) or examine acculturation levels to see how cultural identity and adaptation influence gaming motives and outcomes. Including participants from other host countries or examining other immigrant student groups would test the generalizability of these findings across cultural contexts. Moreover, as gaming culture continues to evolve, research should consider emerging contexts like collegiate esports teams, online streaming communities, and virtual reality platforms. These platforms may introduce new social moderators. For instance, the team-based structure of esports might strengthen social support, or alternatively, competitive stress might create new mental health challenges. Finally, incorporating additional sociocultural moderators in analysis (such as family expectations, academic pressure, or gamer identity strength) could illuminate when gaming is beneficial versus harmful for students. By integrating motivational factors with cultural and social variables, future research can better pinpoint when, for whom, and why gaming is linked to mental health. This newfound knowledge could inform tailored interventions and positive gaming practices across diverse student populations.

4.1. Limitations

Despite the valuable insights, our study has limitations. The reliance on self-report measures may introduce biases such as social desirability and recall bias. The cross-sectional design also limits our ability to establish causal relationships between the investigated variables. Future longitudinal research is needed to better understand the directionality of the associations found in this study.

Additionally, the use of multiple linear regressions for each gaming motive, mental health outcome, and sociocultural factor increases the risk of Type I error, and no correction for multiple comparisons was applied [109]. This could potentially inflate the likelihood of finding significant results by chance. Future studies should consider applying statistical corrections or using more conservative significance thresholds.

Furthermore, the study focused on university students from one Canadian university, likely over-representing international students with comparatively higher resources, thereby limiting the generalizability of the findings to young Asians broadly. University students may have unique experiences and stressors that influence their gaming behaviour and mental health outcomes differently from other age groups or non-student populations. Moreover, cultural differences within the Asian student population, such as differences between international and immigrant students or among various Asian countries,

were not explored in depth and warrant further investigation. University students are also more likely to play video games than the broader adult population [85]; therefore, exposure-related findings may not generalize to older, less educated, or lower-resource groups.

5. Conclusion

This study contributes to our understanding of the complex interplay between gaming behaviour, mental health outcomes, and sociocultural factors among Asian international/immigrant students and White Canadian students. Key findings indicate that Asian students engaged more in recreational and customization gaming, with stronger associations between coping/social gaming motives and anxiety symptoms, suggesting heightened vulnerability due to challenges like acculturative stress, language barriers, and social isolation. The protective role of belongingness in reducing depression was found to be more pronounced for Asian students, emphasizing the need for strong social support. Additionally, cultural factors appear to influence the effectiveness of online versus offline support in mitigating social anxiety and depression. Educational institutions could implement the findings to implement culturally sensitive mental health resources, including peer support groups and workshops, focused on healthy gaming practices, and could improve the curriculum design and student services to better support diverse student populations navigating gaming and acculturation challenges.

CRedit authorship contribution statement

Chris Buchan-Pham: Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Selbi Kurbanova:** Writing – review & editing. **Irene Vitoroulis:** Writing – review & editing, Supervision, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendices.

Table 1
Study variables by ethnicity.

Variable	White	Asian	t or χ^2
<i>Demographic variables</i>			
Age	19.32 (1.52)	20.15 (1.80)	t(213) = -3.87***
Gender (% female)	40.15	45.45	$\chi^2(1) = 0.84$
International student (% yes)	1.46	48.18	$\chi^2(1) = 75.43***$
1st or 2nd gen. immigrant (% yes)	25.55	49.09	$\chi^2(1) = 14.27***$
English spoken at home (% yes)	76.12	37.74	$\chi^2(1) = 34.52***$
At least 5 years in Canada (% yes)	98.54	53.64	$\chi^2(1) = 72.07***$
<i>Mental health variables</i>			
Depression symptoms	19.99 (5.90)	19.88 (5.53)	t(239) = 0.15
Anxiety symptoms	15.30 (5.54)	13.88 (4.47)	t(244) = 2.16*
Loneliness symptoms	12.17 (4.17)	12.65 (3.84)	t(240) = -0.93
Social anxiety symptoms	11.89 (4.64)	12.87 (5.29)	t(231) = 0.46

(continued on next page)

Table 1 (continued)

Variable	White	Asian	t or χ^2
<i>Sociocultural variables</i>			
University belongingness	56.53 (13.88)	54.86 (13.47)	t(236) = 0.96
Online social support	18.19 (4.30)	19.19 (4.40)	t(237) = -2.07*
Offline social support	20.61 (3.18)	19.31 (4.13)	t(222) = 2.67**
Gaming-related norms	24.41 (5.56)	24.07 (5.73)	t(128) = 0.42
<i>Gaming variables</i>			
Problematic gaming	27.07 (7.91)	29.77 (8.22)	t(230) = -2.61**
Gaming for recreation	13.42 (2.47)	13.22 (2.47)	t(232) = 0.75
Gaming for social interaction	10.15 (3.62)	11.05 (3.33)	t(240) = -2.03*
Gaming to cope	11.12 (2.75)	11.74 (2.61)	t(238) = -1.79
Gaming for violence	7.29 (3.40)	8.23 (3.46)	t(232) = -2.13*
Gaming for fantasy	11.23 (3.17)	11.39 (3.03)	t(238) = -0.42
Gaming for cognitive challenges	10.95 (2.44)	11.47 (2.36)	t(237) = -1.71
Gaming for customization	11.89 (2.83)	11.75 (2.84)	t(233) = 0.38
Gaming for competition	12.24 (2.47)	11.92 (2.91)	t(214) = -0.93

*p < 0.05. **p < 0.01. ***p < 0.001.

Table 2

Regression results for interaction effects between ethnicity, gaming behaviour, and sociocultural factors on mental health outcomes.^a

Effect	B	SE	95 % CI	t
<i>Main Effect of Ethnicity on Gaming</i>				
Violent Motives × Ethnicity	1.32	0.54	[0.26, 2.38]	t(233) = 2.45*
Social Motives × Ethnicity	1.32	0.58	[0.19, 2.46]	t(233) = 2.29*
<i>Interaction of Ethnicity × Gaming Motives on Mental Health</i>				
Recreational Motives × Ethnicity on Depression	0.06	0.02	[0.02, 0.10]	t(231) = 2.76**
Coping Motives × Ethnicity on Depression	0.58	0.28	[0.02, 1.13]	t(231) = 2.05*
Recreational Motives × Ethnicity on Loneliness	0.04	0.02	[0.007, 0.06]	t(231) = 2.42*
Coping Gaming Motives × Ethnicity on Loneliness	0.51	0.20	[0.11, 0.91]	t(231) = 2.52*
<i>Ethnicity × Gaming Motives × Sociocultural on Mental Health</i>				
Recreational Motives × Gaming Norms × Ethnicity on Depression	0.007	0.003	[0.001, 0.013]	t(227) = 2.29*
Competition Motives × Belonging × Ethnicity on Depression	0.04	0.02	[0.006, 0.08]	t(227) = 2.31*
Cognitive Motives × Belonging × Ethnicity on Depression	0.05	0.02	[0.005, 0.09]	t(227) = 2.21*
Coping Motives × Belonging × Ethnicity on Depression	-0.04	0.02	[-0.08, -0.007]	t(227) = -2.36*
Violent Motives × Belonging × Ethnicity on Loneliness	0.03	0.01	[0.01, 0.05]	t(227) = 3.00**
Recreational Motives × Gaming Norms × Ethnicity on Social Anxiety	0.006	0.003	[0.001, 0.012]	t(227) = 2.19*
Cognitive Motives × Online Support × Ethnicity on Social Anxiety	-0.008	0.004	[-0.015, -0.001]	t(227) = -2.17*

Note. Positive B values indicate stronger effects for Asian students compared to White students, while negative B values indicate weaker effects for Asian students compared to White students.

*p < 0.05. **p < 0.01. ***p < 0.001.

^a All the variables were centred.

Data availability

Data will be made available on request.

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